

CE 310 — Week 5 Assignment: Normal Distribution Applications

Due: Wed Sep 30, 9:00 AM — submit to D2L Total Points: 60 pts

Overview

This assignment extends the Week 5 in-class exercise into new engineering scenarios. You will apply the Normal distribution parameters you computed in the exercise — concrete compressive strength and vehicle speed — to thresholds and percentiles not covered in the exercise, interpret the results through an engineering design lens, and explain in writing what the distribution reveals that the mean alone cannot.

All three problems use parameters you already computed: `Week5_CE_Measurements.csv`

Do not re-derive μ and σ from scratch — read the values off your completed exercise Answer Sheet and type them into your assignment formulas.

Type the numbers, do not link to the other file. The exercise and the assignment are two separate workbooks. A formula pointing at the exercise file carries a link that breaks the moment either file is renamed or moved, and your grader opens the assignment on its own. Copy the value across.

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Where answers go	Column A of the Answers sheet labels every row — P1b_written, P2a_written, and so on. Each question names its row; type into the shaded cell on that row

Requirement	Detail
Column layout	Enter your formula in Column C and the answer in Column D ; written responses go in Column D of their own labeled row — do not insert, delete, or move columns
Formulas	Type them as text — put an apostrophe before the =, e.g. '=NORM.DIST(5200,4800,250,TRUE)', so Excel stores the formula instead of running it. Screenshots are not accepted
Name and NetID	Type them in the Name and NetID cells at the top of the Answers sheet — that is what matches your work to your student record. A 5-point deduction applies if the NetID is blank.
File name	CE310_W5_Assignment_<NetID>.xlsx (e.g. CE310_W5_Assignment_abc123.xlsx)
Submission	Upload the .xlsx file to D2L → Assignments by 9:00 AM, Wed Sep 30

Problem 1 — New Concrete Strength Probabilities (30 pts)

Use the all-specimen Normal model from your exercise: $\mu = A1$ ($\approx 4,496$ psi), $\sigma = A2$ (≈ 442 psi).

These problems use thresholds different from those in the exercise. Do not reuse exercise thresholds.

Part 1a — High-Strength Classification (10 pts)

In ASTM and industry practice, concrete achieving **5,200 psi or more** is often classified as “high-strength” concrete. Compute:

$P(f_c > 5,200 \text{ psi})$

The fraction of specimens from this mix that would qualify as high-strength.

Show your formula and result. Then write **2–3 sentences** interpreting this value: What does this percentage tell a materials engineer about how reliably their mix

achieves high-strength performance? Include the z-score for 5,200 psi in your interpretation.

Enter your written response in the shaded cell on row P1a_written of the Answers sheet.

Part 1b — Target Zone Probability (10 pts)

A QC engineer wants to know how often batches land in the “target zone” — neither too weak nor over-designed. Compute:

P(4,500 psi < f'_c < 5,000 psi)

Show your formula and result. Then write **2–3 sentences** interpreting this value: What fraction of batches land in this target zone? Is this range wider or narrower than you would expect based on the 68–95–99.7 rule, and why?

Enter your written response in the shaded cell on row P1b_written of the Answers sheet.

Part 1c — Eurocode 7 Characteristic Strength (10 pts)

Eurocode 7 defines the **characteristic value** of a material property as its 5th percentile — the value that 95% of specimens are expected to exceed. Compute:

The 5th percentile concrete compressive strength

Show your formula and result. Then write **2–3 sentences** interpreting this value: How does the 5th percentile compare to the specified design strength of $f'_c = 4,000$ psi? What does this imply about the suitability of the all-specimen mix design for a project governed by the Eurocode 7 characteristic value criterion?

Enter your written response in the shaded cell on row P1c_written of the Answers sheet.

Problem 2 — Traffic Speed Compliance Analysis (18 pts)

This problem uses the three speed Normal models from your exercise. Reference your exercise Answer Sheet cells directly.

- All speeds: $\mu = B1$ (≈ 54.4 mph), $\sigma = B2$ (≈ 7.8 mph)
 - AM Peak: $\mu = B8$ (≈ 52.2 mph), $\sigma = B9$ (≈ 8.2 mph)
 - PM Peak: $\mu = B10$ (≈ 57.2 mph), $\sigma = B11$ (≈ 6.2 mph)
-

Part 2a — Excessive Speed Exceedance (6 pts)

In some federal highway jurisdictions, **65 mph** is the threshold used to flag “excessive speed” for enforcement analysis. Compute $P(\text{Speed} > 65 \text{ mph})$ separately for all three distributions:

- All speeds (use $\mu = B1$, $\sigma = B2$)
- AM Peak only (use $\mu = B8$, $\sigma = B9$)
- PM Peak only (use $\mu = B10$, $\sigma = B11$)

Show all three formulas and results. In **1–2 sentences**, identify which period has the highest exceedance fraction and suggest one reason why that period produces a higher rate than the other, given the same speed limit. Note that the two periods differ in both mean *and* spread — say which of the two is doing more of the work.

Enter your written response in the shaded cell on row P2a_written of the Answers sheet.

Part 2b — 90th Percentile Design Speed (6 pts)

Traffic engineers sometimes use the **90th percentile speed** as a more conservative design speed criterion than the traditional 85th percentile. Compute the 90th percentile for all three distributions:

- All speeds (use $\mu = B1$, $\sigma = B2$)
- AM Peak only (use $\mu = B8$, $\sigma = B9$)
- PM Peak only (use $\mu = B10$, $\sigma = B11$)

Show all three formulas and results. Then write **2–3 sentences**: How do these 90th percentile values compare to the 85th percentile speed you computed in the exercise (B7)? What is the practical difference between using the 85th versus the 90th percentile as the design speed criterion, and which is more conservative?

Enter your written response in the shaded cell on row P2b_written of the Answers sheet.

Part 2c — Evaluating a Speed Limit Argument (6 pts, written response)

A traffic engineer argues:

“The mean speed on this corridor is 54.4 mph — only 9.4 mph over the posted 45 mph limit. A 10 mph speed limit increase to 55 mph would resolve the compliance issue.”

Write **3–4 sentences** that evaluate this argument using your results from P2a and P2b. Does raising the posted limit to 55 mph address the exceedance problem revealed by the 85th percentile? Use at least **two specific numbers** from P2a or P2b in your response.

Enter your written response in the shaded cell on row P2c_written of the Answers sheet.

Problem 3 — Reflection (12 pts)

A project manager reviews the concrete QC report and says:

“Both the Alpha and Beta suppliers show average strength well above 4,000 psi — we don’t need to worry about which supplier we use.”

Write **one paragraph (100–150 words)** that explains why this reasoning is incomplete. Your response must:

1. Reference the ACI 318 10th percentile rule you applied in the exercise
2. Cite the specific 10th percentile values for Alpha ($A_7 \approx 4,191$ psi) and Beta ($A_9 \approx 3,747$ psi)
3. Explain what σ tells you about mix performance that the mean does not

Enter your reflection in the shaded cell on row P3_written of the Answers sheet.

Grading Summary

Problem	Points
P1a — $P(f'_c > 5,200 \text{ psi})$: formula (2) + answer (6) + 2–3 sentence interpretation with z-score (2)	10
P1b — $P(4,500 < f'_c < 5,000)$: formula (2) + answer (6) + 2–3 sentence comparison to 68–95–99.7 rule (2)	10
P1c — 5th percentile strength: formula (2) + answer (6) + 2–3 sentence Eurocode 7 interpretation (2)	10

Problem	Points
P2a — P(Speed > 65 mph) for all / AM / PM: formula (0.5) + answer (1.0) each + 1–2 sentence written response (1.5)	6
P2b — 90th percentile speed for all / AM / PM: formula (0.5) + answer (1.0) each + 2–3 sentence comparison to 85th percentile (1.5)	6
P2c — Speed limit argument evaluation (3–4 sentences, ≥ 2 specific numbers cited)	6
P3 — Reflection paragraph (100–150 words; ACI 318 rule cited; A7 and A9 values cited; σ explained)	12
Total	60 pts